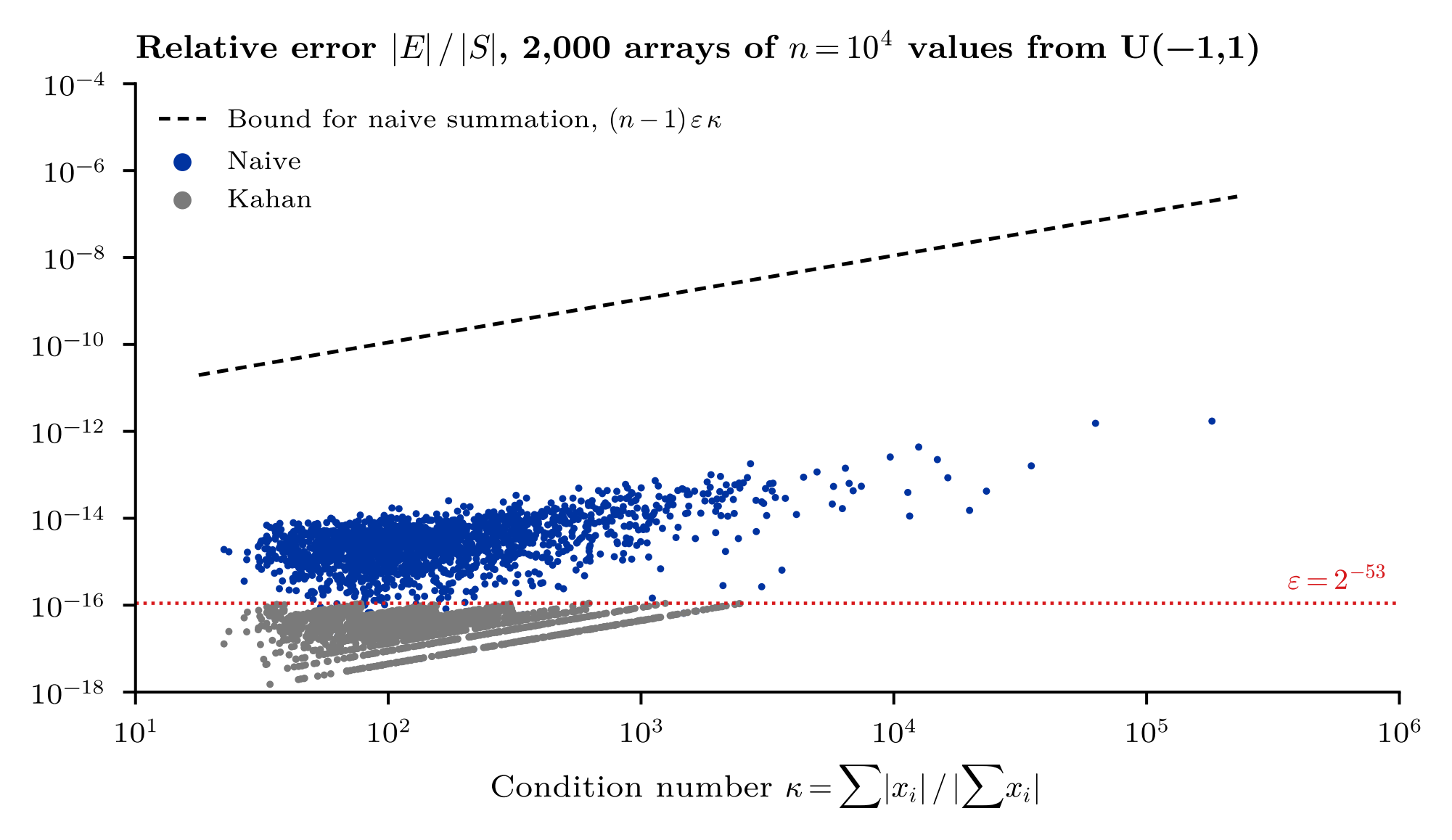

Relative error $|E|\,/\,|S|$, 2,000 arrays of $n = 10^4$ values from U(−1,1)
Bound for naive summation, $(n-1)\,\varepsilon\,\kappa$
Naive
Kahan
$\varepsilon = 2^{-53}$
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
10^{-18}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
Condition number $\kappa = \sum|x_i|\,/\,|\sum x_i|$